ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 10

LIFE SCIENCES

PAPER 1

MARKS: 150 **TIME: 2½ hours**

Topics: chemistry of life, cells, cell division, plant and animal tissues, and the leaf as an organ.

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions. Number your answers correctly and start each question on a new page.
2. Present your answers according to the instructions of each question. Draw diagrams, tables or flow charts only when asked to do so.
3. Diagrams in this question paper are NOT necessarily drawn to scale. You may use a non-programmable calculator.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers. Choose the answer and write only the letter (A–D).

1.1.1 Organic compounds made of carbon, hydrogen and oxygen in the ratio 1 : 2 : 1 are …

A proteins. B lipids.
C carbohydrates. D nucleic acids.

1.1.2 Enzymes are made of …

A carbohydrates. B proteins.
C lipids. D vitamins.

1.1.3 The organelle where aerobic respiration takes place is the …

A ribosome. B mitochondrion.
C chloroplast. D vacuole.

1.1.4 The phase of mitosis in which chromosomes line up on the equator of the cell is …

A prophase. B metaphase.
C anaphase. D telophase.

1.1.5 The plant tissue that transports water from the roots to the leaves is …

A phloem. B xylem.
C epidermis. D parenchyma.

1.1.6 A lack of iron in the diet can cause …

A goitre. B anaemia.
C rickets. D scurvy.

1.1.7 The structure that controls what moves into and out of a cell is the …

A cell wall. B cell membrane.
C nucleus. D cytoplasm.

1.1.8 The movement of water across a selectively permeable membrane, from a high water potential to a low water potential, is …

A diffusion. B osmosis.
C active transport. D phagocytosis.

1.1.9 The animal tissue that is able to contract is …

A epithelial tissue. B connective tissue.
C muscle tissue. D nerve tissue.

1.1.10 A human body cell with 46 chromosomes divides by mitosis. Each daughter cell will have …

A 23 chromosomes. B 46 chromosomes.
C 92 chromosomes. D 12 chromosomes.

(10 × 2) (20)

1.2 Give the correct biological TERM for each of the following descriptions.

1.2.1 Cell division that produces two genetically identical daughter cells

1.2.2 The organelle that contains chlorophyll

1.2.3 Proteins that speed up chemical reactions in living organisms

1.2.4 A group of similar cells that perform the same function

1.2.5 The process in which a cell engulfs solid particles

1.2.6 The building blocks of proteins

1.2.7 Small openings in the epidermis of a leaf that allow gas exchange

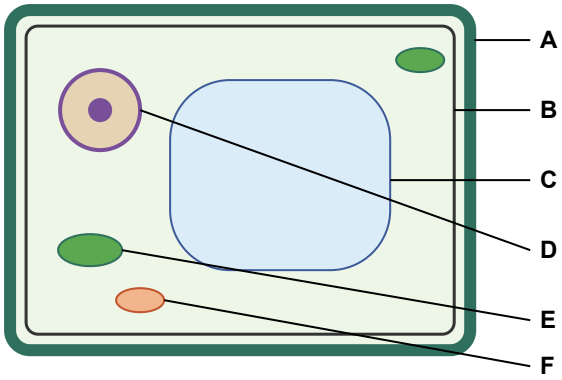
1.2.8 A disease caused by uncontrolled cell division (8 × 1) (8)

1.3 Indicate whether each description in COLUMN I applies to A ONLY, B ONLY, BOTH A AND B, or NONE of the items in COLUMN II.

	COLUMN I	**COLUMN II**
1.3.1	Contains DNA	A: Nucleus B: Mitochondrion
1.3.2	Found in plant cells only	A: Cell wall B: Cell membrane
1.3.3	Transports food (sugars) in plants	A: Xylem B: Phloem
1.3.4	A fat-soluble vitamin	A: Vitamin C B: Vitamin D
1.3.5	Used to test for starch	A: Iodine solution B: Benedict's solution
1.3.6	The stage in which DNA replicates	A: Anaphase B: Telophase

(6 × 2) (12)

1.4 The diagram below shows a plant cell.



1.4.1 Identify parts A to F. (6)

1.4.2 Give the function of part D. (1)

1.4.3 Give the function of part E. (1)

1.4.4 Give the LETTERS of TWO parts that are NOT found in an animal cell. (2)

(10)

TOTAL SECTION A: 50

SECTION B

QUESTION 2

2.1 Learners investigated the effect of temperature on the activity of the enzyme amylase, which breaks down starch. They recorded how long it took for all the starch to be broken down at each temperature.

Temperature (°C)	10	20	30	40	50	60
Time (minutes)	18	10	6	3	8	no reaction

2.1.1 Identify the independent variable and the dependent variable. (2)

2.1.2 At which temperature was the enzyme most active? (1)

2.1.3 Explain why there was no reaction at 60 °C. (2)

2.1.4 Name TWO variables that must be kept constant. (2)

2.1.5 Draw a line graph of the results from 10 °C to 50 °C. (6)

2.1.6 Write a suitable hypothesis for this investigation. (2)

2.1.7 State ONE way to improve the reliability of the results. (1)

(16)

2.2 Food and nutrition.

2.2.1 Name the solution used to test for glucose, and give the colour change for a positive result. (3)

2.2.2 Give TWO reasons why the body needs proteins. (2)

2.2.3 Explain why a diet high in saturated fats is unhealthy. (2)

2.2.4 Name the disease in young children caused by a lack of protein. (1)

(8)

2.3 Two potato pieces, each with a mass of 5,0 g, were placed in different liquids for 30 minutes. Piece X was placed in pure water and its final mass was 5,6 g. Piece Y was placed in a concentrated salt solution and its final mass was 4,1 g.

2.3.1 Calculate the percentage change in mass of each piece. Show your working. (4)

2.3.2 Name the process responsible for the change in mass. (1)

2.3.3 Explain why piece Y lost mass. (3)

(8)

2.4 Tabulate THREE differences between a plant cell and an animal cell. (7)

2.5 Cell division.

2.5.1 List the phases of mitosis in the correct order. (4)

2.5.2 Describe what happens during anaphase. (2)

2.5.3 Give TWO reasons why mitosis is important for an organism. (2)

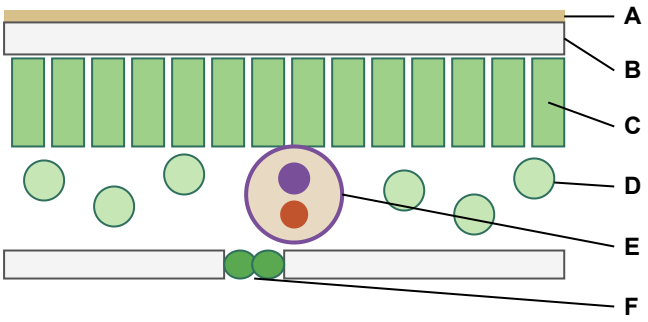
2.5.4 Explain what cancer is, and name ONE method used to treat it. (3)

(11)

[50]

QUESTION 3

3.1 The diagram below shows part of a cross-section through a leaf.



3.1.1 Identify parts A to F. (6)

3.1.2 Give the function of part A and part F. (2)

3.1.3 Explain why the cells of layer C are packed closely together near the upper surface. (2)

(10)

3.2 Animal tissues.

3.2.1 Name the type of tissue that: (a) lines the inside of blood vessels; (b) attaches muscles to bones; (c) carries impulses; (d) covers the outside of the body. (4)

3.2.2 Blood is a connective tissue. Name THREE types of cells or cell fragments found in blood. (3)

(7)

3.3 Plant tissues.

3.3.1 Give TWO differences between xylem and phloem. (4)

3.3.2 Explain how a root hair cell is adapted to absorb water. (3)

(7)

3.4 A learner counted 12 stomata per mm² on the upper epidermis and 48 stomata per mm² on the lower epidermis of a leaf.

3.4.1 Write the ratio of upper to lower stomata in its simplest form. (2)

3.4.2 Explain why most stomata are found on the lower epidermis. (2)

3.4.3 Name the cells that control the opening and closing of a stoma. (1)

(5)

3.5 Stem cells.

3.5.1 What is a stem cell? (2)

3.5.2 Give TWO possible medical uses of stem cells. (2)

3.5.3 Give ONE reason why some people object to the use of embryonic stem cells. (2)

(6)

3.6 Describe the structure of a leaf and explain how it is adapted for photosynthesis and gas exchange. *Content: (12) Synthesis: (3)* (15)

NOTE: NO marks will be awarded for answers in the form of tables, flow charts or diagrams.

[50]

TOTAL SECTION B: 100 · GRAND TOTAL: 150

END OF QUESTION PAPER